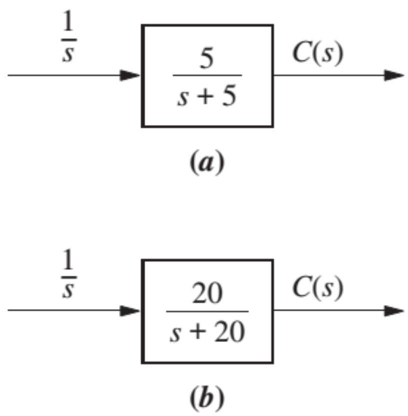
Problem 1:

Find the output response, $c(t)$, for each of the systems shown in Figure. Also find the time constant, rise time, and settling time for each case.



FIGURE

$T_s(2\%) = 4T$

Use the difference between 10% to 90% of y_{ss} as the rise time and also use a 2% settling time.

a) $C(s) = H(s)R(s)$ $\quad H(s) = \frac{5}{s+5} \Rightarrow \boxed{T = \frac{1}{5} = 0.2s}$

$C(s) = \frac{5}{s+5} \cdot \frac{1}{s}$

$C(s) = \frac{5}{s(s+5)} \Rightarrow \int_0^t 5e^{-5\tau}\,d\tau = -e^{-5\tau}\Big|_0^t$

$c(t) = 1 - e^{-5t}$

$c(t) = (1 - e^{-5t})\mu(t)$

$\ln(c(t)) = 5t$

$t = \frac{\ln(c(t))}{5}$

$T_s(2\%) = 4T$

$\boxed{t_s(2\%) = 0.8s}$

$t_R = \frac{1}{5}(\ln(0.9) - \ln(0.1))$

$\boxed{t_R = 0.4394s}$

$C_{ss} = \lim_{y \to \infty} c(t) = 1.$

$10\% C_{ss} = 0.1$

$90\% C_{ss} = 0.9.$

s

b). $C(s) = H(s)R(s)$ $\qquad H(s) = \frac{20}{s+20}$, $\boxed{T = \frac{1}{20}\,s}$

$$C(s) = \frac{20}{s+20} \cdot \frac{1}{s}$$

$$= \frac{20}{s(s+20)} \Rightarrow \int_0^t 20 e^{-20\tau}\, d\tau = 20 \cdot \frac{-1}{20} e^{-20\tau}\Big|_0^t$$

$$c(t) = \left(1 - e^{-20t}\right)\mu(t) \qquad c(t) = 1 - e^{-20t}$$

$$t_s\,(2\%) = 4T$$
$$= 1/5$$

$$c_{ss} = \lim_{t \to \infty} c(t) = 1.$$

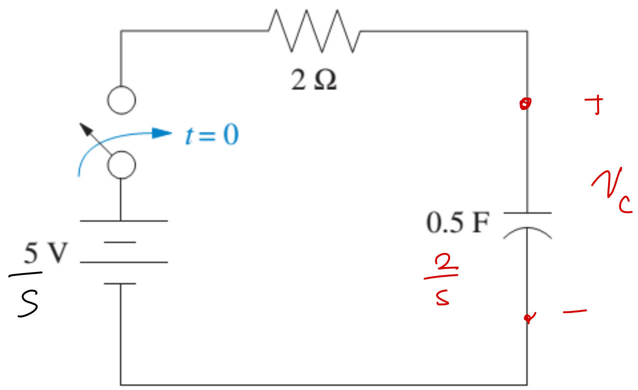
$$\boxed{t_s = 0.2\,s}$$

$$10\%\, c_{ss} = 0.1 \qquad t_R = \frac{1}{20}\ln(9)$$
$$90\%\, c_{ss} = 0.9 \qquad \boxed{t_R = 0.1099\,s}$$

Assignment 9 (ELEC 341 L9_StepResponse)

Problem 2:

Find the capacitor voltage in the network shown in Figure if the switch closes at $t = 0$. Assume zero initial conditions. Also find the time constant, rise time, and settling time for the capacitor voltage.



FIGURE

Use the difference between 10% to 90% of y_{ss} as the rise time and also use a 2% settling time. Use Matlab to verify and plot your output response.

$V_c(s) = V_i(s) \cdot \frac{2/s}{2/s + 2}$ ⇒ $H(s) = \frac{1}{s+1}$

$\boxed{\tau = 1s}$

$V_c(s) = \frac{5}{s} \cdot \frac{2/s}{2/s + 2}$

$= \frac{5}{s} \cdot \frac{2}{2s+2}$

$V_c(s) = \frac{5}{s} \cdot \frac{1}{s+1}$ ⟶ $\int e^{-\tau} d\tau = 1 - e^{-t}$

$\boxed{V_c(t) = 5(1-e^{-t})u(t)}$

$v_{ss} = \lim_{t \to \infty} v(t) = 5V.$

$t_s(2\%) = 4\tau$

$\boxed{t_s(2\%) = 4s}$

$\ln\left(\frac{V_c(t)}{5}\right) = t$

$t_r = \ln\left(\frac{0.9(5)}{5}\right) - \ln\left(\frac{0.1(5)}{5}\right)$

$t_r = \ln(9)$

$\boxed{t_r = 2.1972s}$

Assignment 9 (ELEC 341 L9_StepResponse)

Problem 3:

For the second-order system that follows, find ζ, ω_n, T_s, T_p, T_r, and $\%OS$.

$$T(s) = \frac{16}{s^2+3s+16} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$T = \frac{1}{\zeta\omega_n} = \frac{1}{\frac{3}{8}\cdot\frac{4}{2}} = \frac{2}{3}\,s.$$

$$\omega_n^2 = 16$$

$$\boxed{\omega_n = 4\ rad/s.}$$

$$2\zeta\omega_n = 3$$

$$\zeta = \frac{3}{2\omega_n}$$

$$\boxed{\zeta = \frac{3}{8}}$$

$$\boxed{T_s(2\%) = 4T = 8/3\ s}$$

$$T_p = \frac{\pi}{\omega_d},$$

$$\boxed{T_p = \frac{2\pi}{\sqrt{55}}\ s}$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$= 4\sqrt{1-\frac{9}{64}}$$

$$\omega_d = 4\,\frac{\sqrt{55}}{8}$$

$$\omega_d = \frac{\sqrt{55}}{2}\ rad/s$$

$$\%OS = 100e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \qquad \zeta = 3/8$$

$$\boxed{\%OS = 28.0597\%}$$

$$T_R = \frac{1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1}{\omega_n}$$

$$\boxed{T_R = 0.3559\ s}$$